



Qualification Pack

**Skill India**
कौशल भारता - कुरुता भारता

**THE INSTITUTION OF ENGINEERS (INDIA)**
ESTABLISHED 1920, INCORPORATED BY ROYAL CHARTER 1935

**NCVT**
कौशल गुणवत्ता प्रगति

QUALIFICATION PACK (QP)

ELECTRIC VEHICLE (EV) BATTERY SUSTAINABILITY ENGINEER

Building a Sustainable Future through
Responsible Battery Lifecycle Management

**NSQF LEVEL**
6

**DURATION**
600 HOURS

**AWARD**
CERTIFICATE

**CREDITS**
20

**SECTOR**
AUTOMOTIVE

**COLLECTION & TESTING**

**REFURBISHMENT & REUSE**

**RECYCLING & RESOURCE RECOVERY**

**ENVIRONMENTAL COMPLIANCE**

**CARBON CREDIT & SUSTAINABILITY**

Safe Handling
Quality Assurance

Environmental Compliance
Sustainable Practices

Circular Economy
Greener Tomorrow

Enabling Circular Economy and
Sustainable Mobility for a Better Future

Electric Vehicle (EV) Battery Sustainability Engineer

QP Code: IEI/ASC/Q0101

Version: 1.0

NSQF Level: 6

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IEI/ASC/Q0101: Electric Vehicle (EV) Battery Sustainability Engineer

Brief Job Description

The Electric Vehicle (EV) Battery Sustainability Engineer is responsible for the safe handling, testing, collection, sorting, refurbishment, reuse, recycling, and sustainable disposal of EV batteries. The qualification equips learners with advanced knowledge of battery technologies, hazardous material management, recycling processes, quality assurance, environmental compliance, carbon credit assessment, and circular economy practices to support sustainable growth of the electric mobility sector.

Personal Attributes

The individual should possess analytical and problem-solving abilities, environmental awareness, safety consciousness, attention to detail, technical aptitude, ethical conduct, and communication skills. The candidate should demonstrate teamwork, decision-making capability, adaptability to emerging technologies, and the ability to work responsibly in compliance with quality, safety, and sustainability standards.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [IEI/ASC/N0113: Employability NOS](#)
2. [IEI/ASC/N0111: Battery Types and Technologies](#)
3. [IEI/ASC/N0110: EV Battery Collection and Sorting](#)
4. [IEI/ASC/N0109: Battery Safety and Environmental Practices](#)
5. [IEI/ASC/N0108: Battery Discharge and Preparation](#)
6. [IEI/ASC/N0107: Battery Recycling Techniques and Processes](#)
7. [IEI/ASC/N0106: Post-Recycling Process, Quality Check and Calculation of Carbon Credits](#)
8. [IEI/ASC/N0105: Battery Refurbishment and Reuse](#)
9. [IEI/ASC/N0104: EV Battery Recycling Equipment](#)
10. [IEI/ASC/N0103: Hazardous Materials Handling and Disposal](#)
11. [IEI/ASC/N0102: Reporting, Documentation and Safety Standard](#)
12. [IEI/ASC/N0101: Industry Best Practices and Standard](#)
13. [IEI/ASC/N0112: Introduction to EV Battery Recycling](#)



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Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Electric Vehicles
Occupation	EV Battery Sustainability Engineer
Country	India
NSQF Level	6
Credits	20
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	B.E./B.Tech (Mechanical/ Electrical/ Electronics/ Chemical/ Environmental) with NA of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2030
NSQF Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QC-06-AU-05204-2026-V1-IEI
NQR Version	1



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IEI/ASC/N0113: Employability NOS

Description

This NOS covers the knowledge and skills required to implement industry best practices, maintain workplace documentation, prepare sustainability reports, follow environmental and safety standards, support compliance activities, and maintain operational records in EV battery sustainability and recycling operations.

Scope

The scope covers the following :

- Maintain sustainability and compliance documentation, Follow workplace safety and environmental standards, Prepare operational and sustainability report, Support audit and inspection activities, Maintain hazardous waste records, Implement industry best practices and SOPs

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** communicate effectively with supervisors, team members, customers, and other stakeholders using appropriate verbal and written communication methods.
- PC2.** demonstrate professional behaviour, positive attitude, discipline, and workplace ethics during work activities.
- PC3.** work effectively as an individual and as a team member to achieve organizational objectives.
- PC4.** follow workplace rules, organizational policies, and code of conduct.
- PC5.** apply basic reading, writing, numeracy, and digital literacy skills in workplace activities.
- PC6.** use basic computer applications, internet services, email, and digital platforms for communication and work-related tasks.
- PC7.** identify workplace hazards and follow occupational health, safety, and hygiene practices.
- PC8.** maintain cleanliness, personal hygiene, and proper housekeeping at the workplace
- PC9.** manage time, prioritize tasks, and complete assigned responsibilities efficiently
- PC10.** demonstrate problem-solving, critical thinking, and decision-making abilities in routine workplace situations.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	20	-	-
PC1. communicate effectively with supervisors, team members, customers, and other stakeholders using appropriate verbal and written communication methods.	4	2	-	-
PC2. demonstrate professional behaviour, positive attitude, discipline, and workplace ethics during work activities.	4	2	-	-
PC3. work effectively as an individual and as a team member to achieve organizational objectives.	4	2	-	-
PC4. follow workplace rules, organizational policies, and code of conduct.	4	2	-	-
PC5. apply basic reading, writing, numeracy, and digital literacy skills in workplace activities.	4	2	-	-
PC6. use basic computer applications, internet services, email, and digital platforms for communication and work-related tasks.	4	2	-	-
PC7. identify workplace hazards and follow occupational health, safety, and hygiene practices.	4	2	-	-
PC8. maintain cleanliness, personal hygiene, and proper housekeeping at the workplace	4	2	-	-
PC9. manage time, prioritize tasks, and complete assigned responsibilities efficiently	4	2	-	-
PC10. demonstrate problem-solving, critical thinking, and decision-making abilities in routine workplace situations.	4	2	-	-
NOS Total	40	20	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0113
NOS Name	Employability NOS
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0111: Battery Types and Technologies

Description

This NOS on Battery Types and Technologies provides learners with knowledge and understanding of various EV battery technologies, battery chemistry, charging systems, thermal management, safety practices, applications and sustainability aspects associated with electric vehicle battery systems. The NOS enables learners to identify battery types, follow safe handling procedures and comply with industry standards relevant to EV battery operations.

Scope

The scope covers the following :

- The scope covers identification of battery types, understanding battery technologies, battery specifications, charging systems, thermal management, safety handling procedures and sustainability practices in EV battery applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify different types of EV batteries and their applications.
- PC2.** describe the construction and working principles of various battery chemistries such as Lead-acid, Nickel Metal Hydride (NiMH), Lithium-ion, and emerging battery technologies.
- PC3.** distinguish between different battery cell formats including cylindrical, prismatic, and pouch cells.
- PC4.** explain key battery parameters such as voltage, capacity, energy density, power density, State of Charge (SOC), and State of Health (SOH)
- PC5.** explain battery charging and discharging characteristics and their effect on battery performance and life.
- PC6.** identify factors affecting battery degradation and methods for improving battery life and sustainability.
- PC7.** recognize thermal management requirements and safety precautions associated with different battery technologies.
- PC8.** interpret basic technical specifications, labels, datasheets, and safety information related to batteries.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	10	-	-
PC1. Identify different types of EV batteries and their applications.	4	2	-	-
PC2. describe the construction and working principles of various battery chemistries such as Lead-acid, Nickel Metal Hydride (NiMH), Lithium-ion, and emerging battery technologies.	4	2	-	-
PC3. distinguish between different battery cell formats including cylindrical, prismatic, and pouch cells.	3	2	-	-
PC4. explain key battery parameters such as voltage, capacity, energy density, power density, State of Charge (SOC), and State of Health (SOH)	3	1	-	-
PC5. explain battery charging and discharging characteristics and their effect on battery performance and life.	3	2	-	-
PC6. identify factors affecting battery degradation and methods for improving battery life and sustainability.	1	1	-	-
PC7. recognize thermal management requirements and safety precautions associated with different battery technologies.	1	-	-	-
PC8. interpret basic technical specifications, labels, datasheets, and safety information related to batteries.	1	-	-	-
NOS Total	20	10	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0111
NOS Name	Battery Types and Technologies
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0110: EV Battery Collection and Sorting

Description

This NOS covers the knowledge and skills required for safe collection, segregation, identification, handling, storage, and sorting of end-of-life EV batteries and battery components. It includes battery chemistry identification, safety precautions, hazardous material handling, environmental compliance, documentation practices, and workplace safety procedures followed in EV battery collection and sorting operations.

Scope

The scope covers the following :

- The scope covers identification of battery types, understanding battery technologies, battery specifications, charging systems, thermal management, safety handling procedures and sustainability practices in EV battery applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different categories and types of used EV batteries received for collection and sorting.
- PC2.** inspect EV batteries for visible damage, leakage, swelling, corrosion, or other safety hazards prior to handling.
- PC3.** classify batteries based on chemistry, size, condition, manufacturer specifications, and application type.
- PC4.** segregate batteries according to reusable, repairable, recyclable, and hazardous categories as per prescribed procedures.
- PC5.** verify battery labels, serial numbers, hazard symbols, and technical specifications before sorting activities.
- PC6.** follow standard operating procedures (SOPs) for safe handling, lifting, transportation, and temporary storage of EV batteries.
- PC7.** use appropriate tools, equipment, containers, and personal protective equipment (PPE) during collection and sorting activities.
- PC8.** follow emergency response procedures in case of battery leakage, fire, thermal event, or accidental exposure.
- PC9.** adopt sustainable and environmentally responsible practices during battery collection and sorting operations.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	20	-	-
PC1. identify different categories and types of used EV batteries received for collection and sorting.	3	3	-	-
PC2. inspect EV batteries for visible damage, leakage, swelling, corrosion, or other safety hazards prior to handling.	3	3	-	-
PC3. classify batteries based on chemistry, size, condition, manufacturer specifications, and application type.	3	2	-	-
PC4. segregate batteries according to reusable, repairable, recyclable, and hazardous categories as per prescribed procedures.	1	2	-	-
PC5. verify battery labels, serial numbers, hazard symbols, and technical specifications before sorting activities.	1	2	-	-
PC6. follow standard operating procedures (SOPs) for safe handling, lifting, transportation, and temporary storage of EV batteries.	1	2	-	-
PC7. use appropriate tools, equipment, containers, and personal protective equipment (PPE) during collection and sorting activities.	1	2	-	-
PC8. follow emergency response procedures in case of battery leakage, fire, thermal event, or accidental exposure.	1	2	-	-
PC9. adopt sustainable and environmentally responsible practices during battery collection and sorting operations.	1	2	-	-
NOS Total	15	20	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0110
NOS Name	EV Battery Collection and Sorting
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0109: Battery Safety and Environmental Practices

Description

This NOS covers the competencies required to ensure safe handling, operation, storage, transportation, refurbishment, recycling and disposal of EV batteries in compliance with applicable safety, health and environmental regulations. It includes identification of battery hazards, implementation of workplace safety measures, use of PPE, emergency response procedures, fire prevention practices, hazardous waste management, environmental protection measures and sustainable battery lifecycle management practices.

Scope

The scope covers the following :

- This NOS covers the competencies required to identify battery-related hazards and implement safe working practices during handling, testing, charging, storage, transportation, refurbishment, recycling and disposal of EV batteries. It includes compliance with occupational health, safety and environmental regulations, use of PPE, emergency response procedures, fire prevention measures, hazardous waste management and adherence to sustainability practices applicable to EV battery systems.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazards and risks associated with EV batteries, battery chemicals, electrical systems, and related operations.
- PC2.** follow workplace safety procedures and standard operating procedures (SOPs) during battery handling, storage, transportation, testing, recycling, and disposal activities.
- PC3.** use appropriate personal protective equipment (PPE) and safety gear while performing battery-related tasks.
- PC4.** inspect batteries and work areas for unsafe conditions, damage, leakage, corrosion, overheating, or environmental hazards.
- PC5.** follow safe storage practices for different battery chemistries and hazardous materials.
- PC6.** implement emergency response procedures in case of fire, chemical exposure, electrolyte leakage, explosion, or electrical hazards.
- PC7.** segregate, label, store, and dispose of battery waste and hazardous materials in accordance with environmental regulations and organizational policies.
- PC8.** identify and report unsafe conditions, incidents, accidents, and environmental non-conformities to the concerned authority.
- PC9.** comply with applicable occupational health, safety, environmental, and hazardous waste management regulations.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	30	-	-
PC1. identify hazards and risks associated with EV batteries, battery chemicals, electrical systems, and related operations.	2	3	-	-
PC2. follow workplace safety procedures and standard operating procedures (SOPs) during battery handling, storage, transportation, testing, recycling, and disposal activities.	2	3	-	-
PC3. use appropriate personal protective equipment (PPE) and safety gear while performing battery-related tasks.	3	4	-	-
PC4. inspect batteries and work areas for unsafe conditions, damage, leakage, corrosion, overheating, or environmental hazards.	2	4	-	-
PC5. follow safe storage practices for different battery chemistries and hazardous materials.	2	4	-	-
PC6. implement emergency response procedures in case of fire, chemical exposure, electrolyte leakage, explosion, or electrical hazards.	1	3	-	-
PC7. segregate, label, store, and dispose of battery waste and hazardous materials in accordance with environmental regulations and organizational policies.	1	3	-	-
PC8. identify and report unsafe conditions, incidents, accidents, and environmental non-conformities to the concerned authority.	1	3	-	-
PC9. comply with applicable occupational health, safety, environmental, and hazardous waste management regulations.	1	3	-	-
NOS Total	15	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0109
NOS Name	Battery Safety and Environmental Practices
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0108: Battery Discharge and Preparation

Description

This NOS covers the competencies required for safe discharge, isolation and preparation of EV batteries prior to refurbishment, reuse, recycling or disposal. It includes identification of battery conditions and hazards, implementation of discharge procedures, use of PPE and insulated tools, handling and storage practices, monitoring of battery parameters and compliance with applicable safety and environmental regulations.

Scope

The scope covers the following :

- Collect, identify, sort, segregate, document, and handle EV batteries based on battery chemistry, condition, safety standards, and recycling guidelines while following environmental and workplace safety procedures for sustainable EV battery management.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify EV batteries and battery packs received for discharge and preparation activities.
- PC2.** inspect batteries for physical damage, leakage, swelling, corrosion, overheating, or other safety risks before initiating discharge operations.
- PC3.** verify battery specifications, voltage levels, State of Charge (SOC), and safety status using appropriate tools and instruments.
- PC4.** follow prescribed standard operating procedures (SOPs) for safe battery discharge and preparation activities.
- PC5.** use suitable discharge equipment, insulated tools, instruments, and personal protective equipment (PPE) during operations.
- PC6.** safely discharge batteries to the prescribed voltage or energy level as per technical and safety requirements.
- PC7.** identify and isolate damaged or unsafe batteries requiring special handling or emergency measures.
- PC8.** prepare discharged batteries for dismantling, storage, transportation, recycling, or further processing as per operational requirements.
- PC9.** segregate battery components, accessories, connectors, and associated materials according to approved procedures.
- PC10.** respond appropriately to emergency situations such as thermal runaway, short circuit, fire, chemical leakage, or accidental exposure.
- PC11.** comply with applicable safety, environmental, and occupational health regulations related to battery discharge and preparation activities.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. identify EV batteries and battery packs received for discharge and preparation activities.	3	3	-	-
PC2. inspect batteries for physical damage, leakage, swelling, corrosion, overheating, or other safety risks before initiating discharge operations.	3	3	-	-
PC3. verify battery specifications, voltage levels, State of Charge (SOC), and safety status using appropriate tools and instruments.	3	2	-	-
PC4. follow prescribed standard operating procedures (SOPs) for safe battery discharge and preparation activities.	2	2	-	-
PC5. use suitable discharge equipment, insulated tools, instruments, and personal protective equipment (PPE) during operations.	2	2	-	-
PC6. safely discharge batteries to the prescribed voltage or energy level as per technical and safety requirements.	1	3	-	-
PC7. identify and isolate damaged or unsafe batteries requiring special handling or emergency measures.	1	3	-	-
PC8. prepare discharged batteries for dismantling, storage, transportation, recycling, or further processing as per operational requirements.	1	3	-	-
PC9. segregate battery components, accessories, connectors, and associated materials according to approved procedures.	1	3	-	-
PC10. respond appropriately to emergency situations such as thermal runaway, short circuit, fire, chemical leakage, or accidental exposure.	1	3	-	-
PC11. comply with applicable safety, environmental, and occupational health regulations related to battery discharge and preparation activities.	2	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0108
NOS Name	Battery Discharge and Preparation
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0107: Battery Recycling Techniques and Processes

Description

This NOS enables learners to understand and perform EV battery recycling techniques and processes including battery dismantling, segregation, material recovery, recycling workflows, hazardous material handling, environmental sustainability practices, workplace safety procedures, and compliance with industry regulations related to EV battery recycling operations.

Scope

The scope covers the following :

- Understanding EV battery recycling concepts and workflows. Identifying recyclable battery materials and battery chemistries. Performing battery dismantling, segregation, and safe handling operations. Applying mechanical, hydrometallurgical, and pyrometallurgical recycling techniques. Recovering valuable materials such as lithium, cobalt, nickel, and manganese from used EV batteries. Following workplace safety procedures and hazardous waste management practices. Complying with environmental regulations and sustainability standards related to battery recycling operations. Maintaining documentation, reporting, and operational records for recycling activities.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different battery recycling methods and processes applicable to various EV battery chemistries.
- PC2.** inspect and verify batteries and battery materials before initiating recycling operations.
- PC3.** follow standard operating procedures (SOPs) for dismantling, material separation, recycling, and recovery processes.
- PC4.** use appropriate tools, machinery, equipment, and personal protective equipment (PPE) during recycling activities.
- PC5.** segregate battery components and materials such as metals, plastics, electrolytes, and electronic parts for further processing or recovery.
- PC6.** identify hazardous materials, toxic substances, and contaminated waste generated during recycling operations.
- PC7.** handle, store, and dispose of hazardous waste and by-products in accordance with environmental and safety regulations.
- PC8.** maintain traceability, documentation, and reporting records related to recycling operations and recovered materials.
- PC9.** inspect recycled materials for quality, contamination, and suitability for reuse or further processing.
- PC10.** implement environmental protection measures and pollution control practices during recycling operations.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. identify different battery recycling methods and processes applicable to various EV battery chemistries.	2	3	-	-
PC2. inspect and verify batteries and battery materials before initiating recycling operations.	2	3	-	-
PC3. follow standard operating procedures (SOPs) for dismantling, material separation, recycling, and recovery processes.	2	3	-	-
PC4. use appropriate tools, machinery, equipment, and personal protective equipment (PPE) during recycling activities.	2	3	-	-
PC5. segregate battery components and materials such as metals, plastics, electrolytes, and electronic parts for further processing or recovery.	2	3	-	-
PC6. identify hazardous materials, toxic substances, and contaminated waste generated during recycling operations.	2	3	-	-
PC7. handle, store, and dispose of hazardous waste and by-products in accordance with environmental and safety regulations.	2	3	-	-
PC8. maintain traceability, documentation, and reporting records related to recycling operations and recovered materials.	2	3	-	-
PC9. inspect recycled materials for quality, contamination, and suitability for reuse or further processing.	2	3	-	-
PC10. implement environmental protection measures and pollution control practices during recycling operations.	2	3	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0107
NOS Name	Battery Recycling Techniques and Processes
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



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IEI/ASC/N0106: Post-Recycling Process, Quality Check and Calculation of Carbon Credits

Description

This NOS covers the competencies required to perform post-recycling activities, conduct quality inspection and verification of recovered battery materials, assess recycling efficiency, maintain traceability records, and support the calculation and reporting of carbon emission reductions and carbon credits arising from EV battery recycling and resource recovery operations.

Scope

The scope covers the following :

- The scope covers:
- Post-Recycling Material Inspection and Quality Assessment
- Inspection and verification of recovered materials
- Quality testing and classification of recycled products
- Identification of non-conforming materials
- Recovery Efficiency and Documentation
- Measurement of material recovery rates
- Recording recycling outputs and process data
- Maintenance of traceability and compliance records
- Carbon Credit and Sustainability Assessment
- Collection and analysis of environmental performance data
- Calculation of resource conservation and emission reduction metrics
- Support for carbon credit estimation and sustainability reporting
- Compliance and Environmental Management
- Adherence to quality, environmental, and sustainability standards
- Safe handling of recycled materials
- Promotion of circular economy and resource recovery practices

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** inspect recycled battery materials and recovered components for quality, contamination, and usability.
- PC2.** verify the quantity, purity, and recovery efficiency of extracted materials using prescribed methods and standards.
- PC3.** classify recovered materials based on quality specifications and intended reuse or disposal requirements.
- PC4.** perform testing and quality checks on recycled products, materials, and by-products using appropriate tools and procedures.
- PC5.** document recycling output, recovery rates, waste generation, and process efficiency data accurately.
- PC6.** identify non- conforming materials, process deviations, or quality issues and report them to the concerned authority.



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- PC7.** follow standard operating procedures (SOPs) and quality control guidelines during post-recycling activities
- PC8.** maintain traceability records and documentation related to recycled materials, testing, recovery, and disposal processes.
- PC9.** calculate material recovery percentages, recycling efficiency, and resource conservation metrics using prescribed formulas and data.
- PC10.** collect and compile operational data required for carbon footprint assessment and carbon credit calculations.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. inspect recycled battery materials and recovered components for quality, contamination, and usability.	2	3	-	-
PC2. verify the quantity, purity, and recovery efficiency of extracted materials using prescribed methods and standards.	2	3	-	-
PC3. classify recovered materials based on quality specifications and intended reuse or disposal requirements.	2	3	-	-
PC4. perform testing and quality checks on recycled products, materials, and by-products using appropriate tools and procedures.	2	3	-	-
PC5. document recycling output, recovery rates, waste generation, and process efficiency data accurately.	2	3	-	-
PC6. identify non- conforming materials, process deviations, or quality issues and report them to the concerned authority.	2	3	-	-
PC7. follow standard operating procedures (SOPs) and quality control guidelines during post-recycling activities	2	3	-	-
PC8. maintain traceability records and documentation related to recycled materials, testing, recovery, and disposal processes.	2	3	-	-
PC9. calculate material recovery percentages, recycling efficiency, and resource conservation metrics using prescribed formulas and data.	2	3	-	-
PC10. collect and compile operational data required for carbon footprint assessment and carbon credit calculations.	2	3	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0106
NOS Name	Post-Recycling Process, Quality Check and Calculation of Carbon Credits
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0105: Battery Refurbishment and Reuse

Description

This NOS covers the knowledge, skills and competencies required for refurbishment, repair, testing, grading and reuse of EV batteries. It includes battery diagnostics, safety procedures, performance evaluation, quality checks, lifecycle assessment, reassembly processes and sustainable reuse practices. The NOS also addresses environmental compliance, documentation and operational management associated with extending battery life and enabling circular economy practices within the EV battery ecosystem.

Scope

The scope covers the following :

- Assess battery health, performance parameters, and suitability for refurbishment and reuse.
- Perform battery inspection, testing, diagnosis, and fault identification using appropriate tools and techniques.
- Carry out battery dismantling, repair, replacement of defective components, and reassembly processes.
- Implement battery balancing, conditioning, validation, and performance restoration procedures.
- Apply quality control measures and safety practices during refurbishment operations.
- Ensure compliance with environmental regulations, battery waste management rules, and industry standards.
- Evaluate second-life applications and reuse opportunities for refurbished EV batteries.
- Maintain documentation, traceability records, and reporting related to battery refurbishment and reuse activities.
- Promote circular economy principles through resource optimization, life extension, and sustainable battery management practices.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify EV batteries and battery components suitable for refurbishment and reuse activities.
- PC2.** inspect batteries for physical condition, damage, leakage, swelling, corrosion, and operational safety before refurbishment.
- PC3.** assess battery parameters such as voltage, capacity, State of Charge (SOC), and State of Health (SOH) using appropriate testing methods and instruments.
- PC4.** classify batteries based on refurbishment feasibility, performance condition, and reuse potential
- PC5.** follow standard operating procedures (SOPs) for battery refurbishment, repair, replacement, and reuse activities.
- PC6.** safely dismantle and reassemble battery packs, modules, or cells as per prescribed procedures.
- PC7.** replace defective or degraded battery components using approved techniques and specifications.



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- PC8.** perform testing and validation of refurbished batteries to verify performance, safety, and operational reliability.
- PC9.** identify batteries unsuitable for refurbishment and route them for recycling or safe disposal as per approved procedures.
- PC10.** maintain records and documentation related to battery inspection, refurbishment, testing, reuse, and traceability.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. identify EV batteries and battery components suitable for refurbishment and reuse activities.	2	3	-	-
PC2. inspect batteries for physical condition, damage, leakage, swelling, corrosion, and operational safety before refurbishment.	2	3	-	-
PC3. assess battery parameters such as voltage, capacity, State of Charge (SOC), and State of Health (SOH) using appropriate testing methods and instruments.	2	3	-	-
PC4. classify batteries based on refurbishment feasibility, performance condition, and reuse potential	2	3	-	-
PC5. follow standard operating procedures (SOPs) for battery refurbishment, repair, replacement, and reuse activities.	2	3	-	-
PC6. safely dismantle and reassemble battery packs, modules, or cells as per prescribed procedures.	2	3	-	-
PC7. replace defective or degraded battery components using approved techniques and specifications.	2	3	-	-
PC8. perform testing and validation of refurbished batteries to verify performance, safety, and operational reliability.	2	3	-	-
PC9. identify batteries unsuitable for refurbishment and route them for recycling or safe disposal as per approved procedures.	2	3	-	-
PC10. maintain records and documentation related to battery inspection, refurbishment, testing, reuse, and traceability.	2	3	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0105
NOS Name	Battery Refurbishment and Reuse
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0104: EV Battery Recycling Equipment

Description

This NOS covers the competencies required to identify, operate, inspect, maintain, and safely use EV battery recycling equipment and tools used in dismantling, discharge, segregation, shredding, separation, and recycling processes in compliance with safety and environmental regulations.

Scope

The scope covers the following :

- Identification and operation of EV battery recycling equipment, Safe handling and maintenance of recycling machinery, Usage of discharge units, shredders, separators, and dismantling tools, Monitoring equipment performance and operational safety, Following PPE and emergency safety procedures,
- Maintaining equipment records and preventive maintenance schedules, Compliance with environmental and hazardous waste regulations

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different types of equipment and machinery used in EV battery recycling operations.
- PC2.** inspect recycling equipment, tools, instruments, and safety systems before operation.
- PC3.** follow standard operating procedures (SOPs) and manufacturer guidelines for operating recycling equipment.
- PC4.** use appropriate personal protective equipment (PPE) and safety devices while operating recycling machinery and equipment.
- PC5.** operate battery dismantling, shredding, crushing, separation, discharge, and material recovery equipment safely and efficiently.
- PC6.** monitor equipment parameters, operating conditions, and process performance during recycling operations.
- PC7.** identify abnormal equipment behaviour, malfunctions, overheating, leakage, vibration, or operational hazards during use.
- PC8.** perform routine cleaning, inspection, lubrication, and basic preventive maintenance of recycling equipment as per prescribed procedures.
- PC9.** ensure safe loading, unloading, handling, and feeding of battery materials into recycling equipment.
- PC10.** use measuring instruments, monitoring devices, and control systems for recycling operations.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	20	-	-
PC1. identify different types of equipment and machinery used in EV battery recycling operations.	2	2	-	-
PC2. inspect recycling equipment, tools, instruments, and safety systems before operation.	2	2	-	-
PC3. follow standard operating procedures (SOPs) and manufacturer guidelines for operating recycling equipment.	2	2	-	-
PC4. use appropriate personal protective equipment (PPE) and safety devices while operating recycling machinery and equipment.	2	2	-	-
PC5. operate battery dismantling, shredding, crushing, separation, discharge, and material recovery equipment safely and efficiently.	2	2	-	-
PC6. monitor equipment parameters, operating conditions, and process performance during recycling operations.	2	2	-	-
PC7. identify abnormal equipment behaviour, malfunctions, overheating, leakage, vibration, or operational hazards during use.	2	2	-	-
PC8. perform routine cleaning, inspection, lubrication, and basic preventive maintenance of recycling equipment as per prescribed procedures.	2	2	-	-
PC9. ensure safe loading, unloading, handling, and feeding of battery materials into recycling equipment.	2	2	-	-
PC10. use measuring instruments, monitoring devices, and control systems for recycling operations.	2	2	-	-
NOS Total	20	20	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0104
NOS Name	EV Battery Recycling Equipment
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0103: Hazardous Materials Handling and Disposal

Description

This NOS covers competencies related to safe handling, storage, transportation, segregation, treatment, and disposal of hazardous materials generated during EV battery recycling operations. It includes identification of hazardous substances, use of PPE, emergency response procedures, environmental compliance, and disposal practices as per safety and pollution control regulations.

Scope

The scope covers the following :

- The scope covers identification, handling, storage, segregation, transportation, treatment, and disposal of hazardous materials generated during EV battery recycling operations while ensuring environmental sustainability, workplace safety, and regulatory compliance.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous materials, chemicals, wastes, and substances associated with EV batteries and recycling operations.
- PC2.** interpret hazard labels, safety symbols, material safety data sheets (MSDS/SDS), and handling instructions before handling hazardous materials.
- PC3.** follow standard operating procedures (SOPs) and safety guidelines for handling, storage, transportation, and disposal of hazardous materials.
- PC4.** segregate hazardous and non-hazardous waste materials according to prescribed classification and disposal procedures.
- PC5.** safely collect, package, label, and store hazardous waste in designated containers and storage areas.
- PC6.** prevent leakage, contamination, spillage, fire, explosion, or exposure risks during handling and disposal activities.
- PC7.** ensure environmentally sound disposal of hazardous materials through authorized recyclers, treatment facilities, or disposal agencies.
- PC8.** maintain records, manifests, labels, and documentation related to hazardous material handling, movement, and disposal.
- PC9.** inspect storage areas, containers, and handling practices for compliance with safety and environmental requirements.
- PC10.** comply with applicable occupational health, environmental protection, hazardous waste management, and statutory regulations.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	30	-	-
PC1. identify hazardous materials, chemicals, wastes, and substances associated with EV batteries and recycling operations.	2	3	-	-
PC2. interpret hazard labels, safety symbols, material safety data sheets (MSDS/SDS), and handling instructions before handling hazardous materials.	2	3	-	-
PC3. follow standard operating procedures (SOPs) and safety guidelines for handling, storage, transportation, and disposal of hazardous materials.	2	3	-	-
PC4. segregate hazardous and non-hazardous waste materials according to prescribed classification and disposal procedures.	2	3	-	-
PC5. safely collect, package, label, and store hazardous waste in designated containers and storage areas.	2	3	-	-
PC6. prevent leakage, contamination, spillage, fire, explosion, or exposure risks during handling and disposal activities.	2	3	-	-
PC7. ensure environmentally sound disposal of hazardous materials through authorized recyclers, treatment facilities, or disposal agencies.	2	3	-	-
PC8. maintain records, manifests, labels, and documentation related to hazardous material handling, movement, and disposal.	2	3	-	-
PC9. inspect storage areas, containers, and handling practices for compliance with safety and environmental requirements.	2	3	-	-
PC10. comply with applicable occupational health, environmental protection, hazardous waste management, and statutory regulations.	2	3	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0103
NOS Name	Hazardous Materials Handling and Disposal
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0102: Reporting, Documentation and Safety Standard

Description

This NOS covers the knowledge and skills required for maintaining accurate records, preparing technical and regulatory documentation, and complying with safety standards in EV battery recycling and sustainability operations. It includes documentation of battery collection, storage, transportation, recycling, refurbishment and disposal activities, preparation of compliance reports, incident reporting, maintenance of operational records, and adherence to occupational health, safety, environmental, and statutory requirements.

Scope

The scope covers the following :

- The scope of this NOS includes:
- 1. Maintaining operational records and documentation related to EV battery handling, recycling, refurbishment and disposal.
- 2. Preparing regulatory, environmental and compliance reports as per applicable standards.
- 3. Recording inventory, process parameters, quality inspection results and safety observations.
- 4. Implementing occupational health, safety and environmental protection measures.
- 5. Ensuring proper use of PPE and safe handling procedures for hazardous materials.
- 6. Reporting incidents, near misses, accidents and non-conformities.
- 7. Following emergency preparedness and response procedures.
- 8. Maintaining documentation required for audits, certifications and statutory compliance.
- 9. Adhering to industry standards, environmental regulations and organizational safety policies.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** • maintain accurate records and documentation related to battery handling, recycling, refurbishment, testing, storage, transportation, and disposal activities.
- PC2.** record operational data, inspection findings, testing results, material movement, and process performance as per prescribed formats and procedures.
- PC3.** prepare and update reports related to safety incidents, equipment status, quality checks, hazardous materials, and operational activities.
- PC4.** maintain traceability and identification records for batteries, recycled materials, recovered components, and hazardous waste.
- PC5.** verify and update labels, tags, manifests, logbooks, and compliance documents related to battery operations.
- PC6.** follow standard operating procedures (SOPs), reporting protocols, and documentation requirements prescribed by the organization and regulatory authorities.
- PC7.** ensure confidentiality, accuracy, and integrity of operational records and reports.
- PC8.** identify and report unsafe conditions, non-conformities, equipment malfunctions, environmental hazards, and process deviations to the concerned authority.



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- PC9.** comply with occupational health, safety, environmental protection, and hazardous waste management standards during all operational activities.
- PC10.** use appropriate personal protective equipment (PPE), safety signs, emergency systems, and documentation tools while performing assigned tasks.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	10	-	-
PC1. - maintain accurate records and documentation related to battery handling, recycling, - refurbishment, testing, storage, transportation, and disposal activities.	2	1	-	-
PC2. record operational data, inspection findings, testing results, material movement, and process performance as per prescribed formats and procedures.	2	1	-	-
PC3. prepare and update reports related to safety incidents, equipment status, quality checks, hazardous materials, and operational activities.	2	1	-	-
PC4. maintain traceability and identification records for batteries, recycled materials, recovered components, and hazardous waste.	2	1	-	-
PC5. verify and update labels, tags, manifests, logbooks, and compliance documents related to battery operations.	2	1	-	-
PC6. follow standard operating procedures (SOPs), reporting protocols, and documentation requirements prescribed by the organization and regulatory authorities.	2	1	-	-
PC7. ensure confidentiality, accuracy, and integrity of operational records and reports.	2	1	-	-
PC8. identify and report unsafe conditions, non-conformities, equipment malfunctions, environmental hazards, and process deviations to the concerned authority.	2	1	-	-
PC9. comply with occupational health, safety, environmental protection, and hazardous waste management standards during all operational activities.	2	1	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. use appropriate personal protective equipment (PPE), safety signs, emergency systems, and documentation tools while performing assigned tasks.	2	1	-	-
NOS Total	20	10	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0102
NOS Name	Reporting, Documentation and Safety Standard
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0101: Industry Best Practices and Standard

Description

This NOS covers the application of industry best practices, standards, codes, guidelines, and regulatory requirements related to electric vehicle battery sustainability, recycling, refurbishment, reuse, environmental compliance, occupational safety, quality management, and circular economy principles. It enables learners to adopt standardized procedures and benchmark practices for efficient, safe, sustainable, and legally compliant EV battery lifecycle management.

Scope

The scope covers the following :

- Understanding national and international standards applicable to EV batteries and battery recycling operations.
- Applying industry best practices for battery collection, storage, transportation, dismantling, refurbishment, recycling, and disposal.
- Ensuring compliance with Battery Waste Management Rules, environmental regulations, and occupational health & safety requirements.
- Implementing quality assurance and quality control practices throughout battery sustainability operations.
- Adopting circular economy principles for resource recovery, material reuse, and waste minimization.
- Benchmarking operational performance against industry standards and sustainability indicators.
- Promoting continuous improvement, innovation, and responsible engineering practices in EV battery sustainability management.
- Maintaining ethical, safe, and environmentally responsible work practices aligned with organizational and statutory requirements.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify applicable industry standards, regulations, codes, and guidelines related to EV batteries, recycling, safety, and sustainability.
- PC2.** follow organizational policies, standard operating procedures (SOPs), and industry best practices during battery handling, recycling, refurbishment, and disposal activities.
- PC3.** Comply with occupational health, safety, environmental protection, and hazardous waste management requirements.
- PC4.** adopt safe, efficient, and sustainable work practices in battery- related operations.
- PC5.** use approved tools, equipment, testing methods, and technologies in accordance with industry standards and manufacturer guidelines.
- PC6.** identify operational risks, non- conformities, unsafe practices, and process deviations during work activities.
- PC7.** maintain proper documentation, traceability, records, reports, and operational data as per organizational and regulatory requirements.



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- PC8.** follow best practices for energy conservation, resource optimization, waste minimization, and pollution prevention.
- PC9.** implement housekeeping, cleanliness, and workplace organization practices in operational areas.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	10	-	-
PC1. identify applicable industry standards, regulations, codes, and guidelines related to EV batteries, recycling, safety, and sustainability.	2	1	-	-
PC2. follow organizational policies, standard operating procedures (SOPs), and industry best practices during battery handling, recycling, refurbishment, and disposal activities.	2	1	-	-
PC3. Comply with occupational health, safety, environmental protection, and hazardous waste management requirements.	2	1	-	-
PC4. adopt safe, efficient, and sustainable work practices in battery- related operations.	2	1	-	-
PC5. use approved tools, equipment, testing methods, and technologies in accordance with industry standards and manufacturer guidelines.	2	1	-	-
PC6. identify operational risks, non- conformities, unsafe practices, and process deviations during work activities.	2	1	-	-
PC7. maintain proper documentation, traceability, records, reports, and operational data as per organizational and regulatory requirements.	4	2	-	-
PC8. follow best practices for energy conservation, resource optimization, waste minimization, and pollution prevention.	2	1	-	-
PC9. implement housekeeping, cleanliness, and workplace organization practices in operational areas.	2	1	-	-
NOS Total	20	10	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0101
NOS Name	Industry Best Practices and Standard
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025



Qualification Pack

IEI/ASC/N0112: Introduction to EV Battery Recycling

Description

This NOS covers the foundational knowledge, skills, and competencies required for understanding Electric Vehicle (EV) battery recycling processes, battery types, recycling workflow, environmental sustainability practices, hazardous material awareness, and workplace safety procedures. It enables learners to understand safe handling, collection, sorting, dismantling, and recycling concepts associated with EV batteries while supporting sustainable waste management and circular economy practices.

Scope

The scope covers the following :

- This NOS covers the foundational understanding of Electric Vehicle (EV) battery recycling ecosystem, including battery types, composition, lifecycle, recycling processes, environmental sustainability principles, hazardous material awareness, workplace health and safety practices, regulatory compliance requirements, and basic industry best practices associated with EV battery sustainability and recycling operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify different types of EV batteries and their applications.
- PC2.** Explain the battery lifecycle and recycling process flow.
- PC3.** Follow workplace safety procedures while handling EV batteries.
- PC4.** Identify hazardous materials and environmental risks associated with battery recycling
- PC5.** Demonstrate awareness of battery collection, sorting, and dismantling processes
- PC6.** Follow sustainability and waste management practices in recycling operations.
- PC7.** Maintain basic documentation and reporting related to recycling activities
- PC8.** Comply with industry standards, environmental regulations, and safety guidelines



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	10	-	-
PC1. Identify different types of EV batteries and their applications.	4	2	-	-
PC2. Explain the battery lifecycle and recycling process flow.	4	2	-	-
PC3. Follow workplace safety procedures while handling EV batteries.	3	2	-	-
PC4. Identify hazardous materials and environmental risks associated with battery recycling	3	1	-	-
PC5. Demonstrate awareness of battery collection, sorting, and dismantling processes	3	2	-	-
PC6. Follow sustainability and waste management practices in recycling operations.	1	1	-	-
PC7. Maintain basic documentation and reporting related to recycling activities	1	-	-	-
PC8. Comply with industry standards, environmental regulations, and safety guidelines	1	-	-	-
NOS Total	20	10	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	IEI/ASC/N0112
NOS Name	Introduction to EV Battery Recycling
Sector	Automotive
Sub-Sector	
Occupation	EV Battery Sustainability Engineer
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2030
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment shall be conducted covering theory & practical as per NOS. Batches assigned via SIP/email. ToA-certified assessors conduct assessment. Centre must be equipped. Batches with strength more than 30 require 2 assessors. Proper time allocation for theory & practical ensured. Question bank prepared & validated by SMEs and mapped to NOS; assessors ToA and trainers ToT certified. Evidence includes time-stamped, geotagged assessor reports and centre photographs with branding. Surprise inspection may be conducted for verification. All records (attendance, answer sheets, evidences) maintained in hard copy/digital format. OJT assessed includes workplace performance, tool handling, customer interaction & soft skills; video/photo evidence required. Out of the total 20 Credits, 3 Credits as per NCrf have been reserved for OJT.

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage



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Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
IEI/ASC/N0113.Employability NOS	40	20	0	0	60	12
IEI/ASC/N0111.Battery Types and Technologies	20	10	0	0	30	3
IEI/ASC/N0110.EV Battery Collection and Sorting	15	20	0	0	35	3
IEI/ASC/N0109.Battery Safety and Environmental Practices	15	30	0	0	45	4
IEI/ASC/N0108.Battery Discharge and Preparation	20	30	0	0	50	10
IEI/ASC/N0107.Battery Recycling Techniques and Processes	20	30	0	0	50	10
IEI/ASC/N0106.Post-Recycling Process, Quality Check and Calculation of Carbon Credits	20	30	0	0	50	10
IEI/ASC/N0105.Battery Refurbishment and Reuse	20	30	0	0	50	10
IEI/ASC/N0104.EV Battery Recycling Equipment	20	20	0	0	40	10
IEI/ASC/N0103.Hazardous Materials Handling and Disposal	20	30	0	0	50	10
IEI/ASC/N0102.Reporting, Documentation and Safety Standard	20	10	0	0	30	10
IEI/ASC/N0101.Industry Best Practices and Standard	20	10	0	0	30	5
IEI/ASC/N0112.Introduction to EV Battery Recycling	20	10	0	0	30	3
Total	270	280	-	-	550	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.